ELE 2136

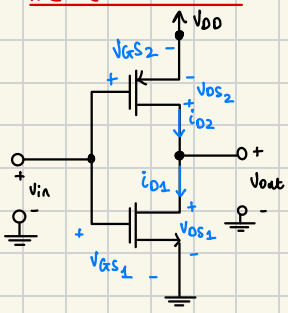
Lecture 9

CMOS

Regions of VTC for the CMOS inverter

Region #	Mode of NMOS (1)	Mode of PMOS (2)
1	Cut-off	Triode
2	Saturation	Triode
3	Saturation	Triode/Saturation
4	Triode	Saturation
5	Triode	Cut-off

The CMOS inverter:



Assumption 1: $V_{tn} = V_t$

$V_{tp} = -V_t$

$V_{tn} = |V_{tp}| = V_t$

Assumption 2: $\mu_n C_{ox}\left(\frac{W}{L}\right)_n = \mu_p C_{ox}\left(\frac{W}{L}\right)_p$

Assumption 3: $\lambda_n = \lambda_p = 0$

Assumption 4: $V_t < V_{DD}$

V_{OH}: MAX Output

V_{OL}: MIN Output

V_{IL}: Input voltage at the beginning of undefined region.

V_{IH}: Input voltage at the end of undefined region.

$NM_H = V_{OH} - V_{IH}$

$NM_L = V_{OL} - V_{IL}$

Ideal case: $NM_H = NM_L$

$= \frac{V_{OH}}{2}$

$= \frac{\text{Supply Voltage}}{2}$

Fact-1: $V_{in} = V_{GS1}$

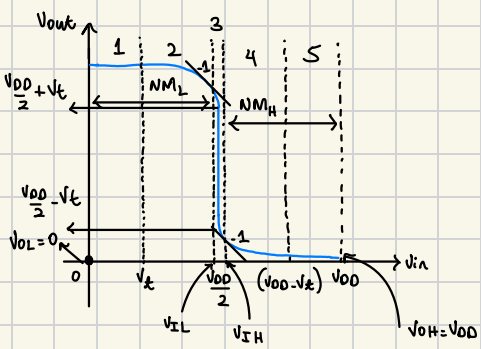
Fact-2: $V_{in} = V_{GS2} + V_{DD}$

Fact-3: $V_{out} = V_{DS1}$

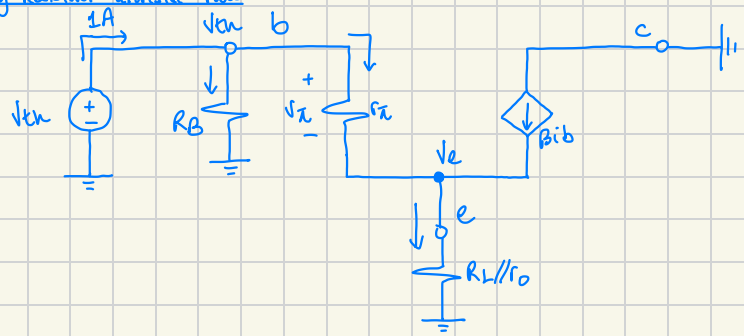
Fact-4: $V_{out} = V_{DS2} + V_{DD}$

Fact-5: $I_{D1} = I_{D2}$

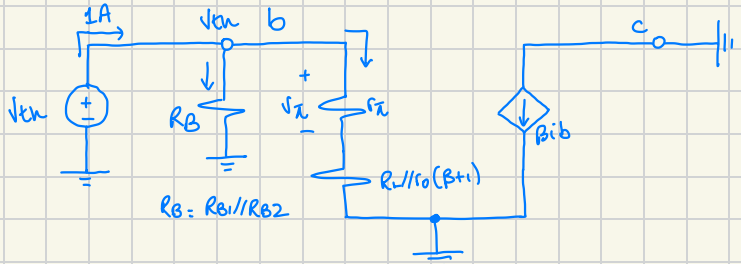
1) $0 < V_{in} < V_t$
2) $V_t < V_{in} < \frac{V_{DD}}{2}$
3) $V_{in} = \frac{V_{DD}}{2}$
4) $\frac{V_{DD}}{2} < V_{in} < V_{DD} - V_t$
5) $V_{DD} - V_t < V_{DD}$



Using Resistor transfer rule

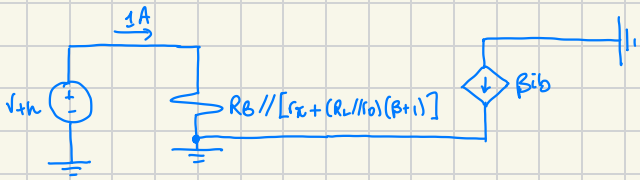


Resistor transfer rule



$$R_B \,//\, [r_\pi + (R_L // r_o)(\beta+1)]$$

Simplifying



$$V_{th} = R_{in} = \underbrace{(R_{B1} // R_{B2})}_{R_B} // [r_\pi + (R_L // r_o)(\beta+1)]$$